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Evaporative thermal management systems and methods

Abstract

Devices and methods are provided herein useful to thermal management. In some embodiments, a thermal management device includes a housing with a fixed amount of working fluid disposed therein. The substrate is in thermal communication with the thermal management device such that evaporation of the working fluid controls the temperature of the substrate. Evaporated working fluid exits the housing through one or more vents. The housing further includes a plurality of supports that increase the surface area to volume ratio of the housing. The high surface area to volume ratio of the housing increases the rate of heat transfer and also minimizes or otherwise reduces the size and weight of the thermal management device. The supports may further serve to mechanically support the substrate, enabling the housing to act as a combined thermal and mechanical device.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2925722	12/1959	Blackburn	N/A	N/A
3066499	12/1961	Fisher	174/15.1	F25B 23/006
5349831	12/1993	Daikoku	N/A	N/A
5398519	12/1994	Weber	N/A	N/A
6533029	12/2002	Phillips	165/104.33	F28D 15/046
6994151	12/2005	Zhou	N/A	N/A
9835363	12/2016	Xiao	N/A	N/A
10515871	12/2018	Joshi	N/A	N/A
10746479	12/2019	Rush	N/A	N/A
2014/0268572	12/2013	Ranjan	N/A	N/A
2019/0021188	12/2018	Phan	N/A	N/A
2019/0166706	12/2018	Trotman	N/A	H05K 5/0214
2019/0170406	12/2018	Wilson	N/A	F25B 19/00
2020/0350229	12/2019	Chang	N/A	N/A
2023/0038664	12/2022	Rush	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3218929	12/2016	EP	N/A
200438318	12/2003	WO	N/A
201651569	12/2015	WO	N/A

OTHER PUBLICATIONS

USPTO; U.S. Appl. No. 17/397,475; Non-Final Rejection mailed Nov. 20, 2023; (pp. 1-13). cited by applicant

USPTO; U.S. Appl. No. 17/397,475; Notice of Allowance and Fees Due (PTOL-85) mailed Feb. 23, 2024; (pp. 1-9). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/397,475, filed on Aug. 9, 2021, which is hereby incorporated herein by reference.

TECHNICAL FIELD

(1) This technical field relates generally to thermal management systems and, more specifically, to thermal management systems for applications requiring low or optimized size, weight, and power.

BACKGROUND

(2) Thermal management systems may employ forced and natural convection, conduction, or radiation to transfer heat to or from a substrate. Passive thermal management systems typically rely on additional thermal mass, for example, in the form of a working fluid, which adds physically mass and size, to control temperatures. In addition, active thermal management systems may rely on additional equipment such as pumps, heat exchangers, or condensers to circulate a working fluid. The addition of mass may be a challenge for applications that require low or optimized size, weight, and power (SWaP).

(3) Therefore, there remains a desire to have a passive thermal management system that efficiently controls the temperature of a substrate while minimizing the size, weight, and power of the system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an embodiment of an evaporative thermal management device having a top substrate configuration.

(2) FIG. 2 is a schematic diagram of an embodiment of an evaporative thermal management device having a bottom substrate configuration.

(3) FIG. 3 is a schematic diagram of an embodiment of an evaporative thermal management device having a top and bottom substrate configuration.

(4) FIG. 4 is a schematic diagram of an embodiment of an evaporative thermal management device including a porous media domain.

(5) FIG. 5 is a schematic diagram of another embodiment of an evaporative thermal management device including a porous media domain.

(6) FIG. 6 is a schematic diagram of a two-piece housing of an evaporative thermal management device.

(7) FIG. 7 is a schematic diagram of a housing of an evaporative thermal management device having branched supports.

(8) FIG. 8 is a perspective view of an embodiment of an evaporative thermal management device.

(9) FIG. 9 is a breakaway view of the evaporative thermal management device of FIG. 8 taken along line 9-9.

(10) FIG. 10 is a schematic diagram of an embodiment of an evaporative thermal management device equipped with vent tubes.

(11) FIG. 11 is a schematic diagram of an embodiment of an evaporative thermal management device including a working liquid held in a porous media domain.

(12) FIG. 12 is schematic diagram of another embodiment of an evaporative thermal management device including a working liquid held in a porous media domain.

(13) FIG. 13 is a schematic diagram of another embodiment of an evaporative thermal management device including a vapor-liquid separator.

(14) FIG. 14 is a schematic diagram of another embodiment of an evaporative thermal management

device including a working liquid held in a porous media domain.

(15) FIG. **15** is a schematic diagram of an embodiment of an evaporative thermal management device equipped with a pressure relief valve.

(16) FIG. **16** is a schematic diagram of an embodiment of an evaporative thermal management device equipped with a venturi pump.

(17) FIG. **17** is a schematic diagram of an embodiment of an evaporative thermal management device having a non-planar housing.

(18) FIG. **18** is a schematic diagram of an embodiment of an evaporative thermal management device having an angled housing.

(19) FIG. **19** is a schematic diagram of an embodiment of an evaporative thermal management device coupled to a secondary substrate.

(20) Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to aid in understanding various embodiments. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted to facilitate a less obstructed view of these various embodiments.

DETAILED DESCRIPTION

(21) With reference to FIGS. **1-5**, there is illustrated an evaporative thermal management device **10** that controls the temperature of a substrate **12**, which may be a heat source, while minimizing or otherwise reducing the device's size and weight. The evaporative thermal management device **10** includes a housing **14** with a working fluid **30** disposed therein. The substrate **12** is in thermal communication with the evaporative thermal management device **10** such that evaporation of the working fluid **30** controls the temperature of the substrate **12**. Evaporated working fluid **30** exits the housing **14** through one or more vents **28**. The housing **14** further includes a plurality of supports **20**. The plurality of supports **20** increase the surface area to volume ratio of the housing **14**. The high surface area to volume ratio of the housing **14** increases the rate of heat transfer and also minimizes or otherwise reduces the size and weight of the evaporative thermal management device **10** by minimizing or otherwise reducing the amount of working fluid **30** required for heat transfer. The supports **20** may further serve to mechanically stiffen the housing **14** and support the substrate **12**, enabling the housing **14** to act as a combined thermal and mechanical device.

(22) Turning back to FIG. **1**, the evaporative thermal management device **10** includes the housing **14**. The housing **14** may be formed in any shape that is complementary to the shape or geometry of the substrate **12**. The housing **14** may also be formed into three-dimensional (3D) shapes to conform to a particular space or volume and to form specific functions. For example, the housing **14** or portions thereof may be formed into a narrow conduit to serve as a vent tube or a vapor liquid separator. In some forms, the housing **14** may include an opening disposed therethrough that is configured to receive one or more substrates **12**. In other forms, the housing **14** may be configured to receive a substrate on its top surface and/or on its bottom surface. The housing **14** includes a base case **16** and a top case **18**. The top case **18** is spaced from and opposite the base case **16**. The cases **16**, **18** may be formed from sheets of material in any suitable shape or configuration.

(23) The housing **14** further includes the plurality of supports **20**, which extend from the base case **16** to the top case **18** and form a cavity **22** therebetween. The cavity **22** is bounded by walls **26** and a base **24**. The supports **20** may be formed in a variety of geometries and arrangements. In some approaches, the geometry and arrangement of the supports **20** may be designed to mechanically stiffen the housing **14** and/or to support a particular substrate **12**. Accordingly, the housing **14** may act as both a thermal device and a support structure for the substrate **12**. That is, the housing **14** may both facilitate heat transfer and bear the weight of the substrate **12**. The supports **20** depicted in FIGS. **1-5** are cylindrical columns. However, the supports **20** may be any suitable shape such as rectangular columns, triangular columns, branched structures, I-beams, L-shaped beams, trihedron

structures, etc. The cross-sectional area of the supports **20** may be uniform or non-uniform.

(24) The working fluid **30** is disposed in the cavity **22** of the housing **14**. The working fluid **30** may be any suitable heat transfer fluid. Suitable working fluids **30** may include water, alcohols (e.g., methanol, ethanol, propanol) as well as mixtures thereof. Other suitable working fluids **30** include other fluids having a high latent heat of vaporization including, for example, ethylene glycol, glycerin, ammonia, or acetone. Working fluids **30** with a higher boiling point, such as glycerin, may be more suitable for applications having a higher threshold temperature. Working fluids **30** with a lower boiling point, such as ammonia, may be more suitable for applications having a lower threshold temperature. In some embodiments, the working fluid **30** may include buffers or other additives.

(25) The boiling point or saturation temperature ($T_{sub.sat}$) of the working fluid **30** in the evaporative thermal management device **10** may be controlled by the type of working fluid **30** and the operating pressure of the evaporative thermal management device **10**, that is, a pressure at which the evaporative thermal management device **10** operates. In some approaches, the working fluid **30** may be selected based, at least in part, on a threshold temperature associated with the substrate **12**. The threshold temperature may be a desired working temperature for the substrate **12**. In some approaches, the evaporative thermal management device **10** may be designed to keep the substrate at or below the threshold temperature. In operation, boiling and evaporation of the working fluid **30** may maintain the substrate **12** at or below the threshold temperature. Accordingly, a particular working fluid **30** may be selected to achieve the threshold temperature for the substrate **12** at a particular operating pressure of the evaporative thermal management device **10**. In some aspects, the working fluid **30** has a boiling point (i.e., saturation temperature) at or below the threshold temperature at the operating pressure of the device.

(26) In some approaches, the amount of working fluid **30** disposed in the housing **14** may achieve a wetted surface area to volume ratio sufficient to dissipate a particular heat load for a particular duration of time. In some approaches, the wetted surface area to volume ratio of the housing **14** may be greater than 100 l/m and, in some aspects, may be in the range of about 100 to 10,000 l/m.

(27) The housing **14** also includes one or more vents **28**. The vents **28** may be separate from or integral with the housing **14**; for example, the housing **14** may morph into the vents **28**. The vents **28** may be in fluid communication with the ambient environment surrounding the housing **14**. The vents **28** act exit ports through which evaporated working fluid **30** may exit the evaporative thermal management device **10**. The vents **28** may be formed in either the top case **18** or the base case **16** of the housing **14**. In FIG. 2, the vents **28** are shown in an open configuration, allowing evaporated working fluid **30** to freely exit the cavity **22**. However, it is also contemplated that the vents **28** may be sealed. The vents **28** may be sealed, for example, for storage purposes before the evaporative thermal management device **10** is placed in operation. In some embodiments, the vents **28** may be sealed with a valve such as a squib valve (not shown) which may be actuated to allow evaporated working fluid to flow out of the cavity **22** to the ambient environment **34**. While the vents **28** are included on the top case **18** in FIG. 1, it is contemplated that the vents **28** may be disposed on any portion of the housing **14**. In some approaches, the position of the vents **28** may depend on the orientation of the evaporative thermal management device **10** during operation. For example, the vents **28** may be positioned to prevent working fluid **30** that is in the liquid state from flowing out of the housing **14** in a particular orientation.

(28) In some embodiments, the vents **28** expose the cavity **22** of the evaporative thermal management device **10** to a pressure of the ambient environment **34**. Thus, the operating pressure of the evaporative thermal management device **10** may be the pressure of the ambient environment **34**. In other embodiments, the vents **28** may be pressure regulated. To achieve pressure regulation, the vents **28** may be equipped with pressure regulating devices (such as pressure relief valve **92** and venturi pump **94**) for example, as shown in FIGS. 14 and 15. The pressure regulating devices may control the operating pressure of the evaporative thermal management device **10**. The vents **28** may

be used to expose the working fluid to the targeted saturation pressure ($P_{sub.sat}$), for example, that yields a saturation temperature equal to or less than the threshold temperature for the substrate **12**.

(29) The evaporative thermal management device **10** is in thermal communication with one or more substrates **12**. The substrate **12** may be in thermal communication with any portion of the evaporative thermal management device **10** so that heat may be transferred to and/or from the evaporative thermal management device **10** and the substrate **12**. In some embodiments, the substrate **12** is a heat source or heat producing device. The substrate **12** may be, for example, an electronic device such as a silicon device, a microprocessor, an integrated circuit, or a semiconductor; however, it should be understood that the evaporative thermal management device **10** is not limited to use with such substrates. In some approaches, the substrate **12** may be an actuator, a battery (e.g., a thermal battery), a sensor (e.g., a navigation sensor, a mission sensor), or other temperature sensitive payloads.

(30) In operation, the cavity **22** of the evaporative thermal management device **10** is filled with a fixed amount of working fluid **30**. The working fluid **30** is heated by the substrate **12**. The transfer of heat from the substrate **12** to the working fluid **30** cools the substrate **12** and causes the working fluid **30** to boil and evaporate. This boiling of the working fluid **30** maintains the substrate **12** at or near the boiling point temperature of the working fluid **30**. In some approaches, the boiling point temperature of the working fluid **30** may be equal to or less than the threshold temperature for the substrate **12**. The heated working fluid **30** boils and forms a vapor within the cavity **22**. Evaporated working fluid **30** exists the cavity **22** of the evaporative thermal management device **10** through the vents **28**. Evaporated working fluid **30** passes through the vents **28** to the ambient environment **34**. When the evaporative thermal management device **10** is at ambient pressure (i.e., when vents **28** are not pressure regulated), the evaporative thermal management device **10** may employ a working fluid **30** having a saturation pressure at or below the ambient pressure when the evaporative thermal management device **10** is at or below the threshold temperature.

(31) In some embodiments, when the vents **28** are pressure regulated (see FIGS. **14** and **15**), pressure regulating devices (such as pressure relief valve **92** or venturi pump **94**) control the operating pressure of the evaporative thermal management device **10**. The pressure regulating devices may adjust the operating pressure of the evaporative thermal management device **10** to be at or below the saturation pressure of the working fluid **30**, for example, at the threshold temperature. In this manner, the operating pressure of the evaporative thermal management device **10** controls the boiling point of the working fluid **30**.

(32) In operation, the evaporative thermal management device **10** is charged with a fixed amount of working fluid **30**. The evaporative thermal management device **10** dissipates heat through evaporation of the working fluid **30** rather than through circulation of the working fluid **30**. Accordingly, the evaporative thermal management device **10** does not require equipment such as pumps, secondary heat exchangers, and condensers for the working fluid **30**. Because such equipment is not required, the weight of the thermal management is minimized.

(33) Including the supports **20** in the housing **14** increases the surface area of the housing **14** that is in contact with the working fluid **30** and, accordingly, increases the rate of heat transfer. In some approaches, the housing **14** and supports **20** may be configured to provide a particular surface area in order to achieve a desired rate of heat transfer.

(34) Because the evaporative thermal management device **10** is charged with a fixed amount of working fluid, the evaporative thermal management device **10** may be designed for operation for a predetermined period of time. The evaporative thermal management device **10** may be designed for a single use or multiple uses. In some embodiments, when the evaporative thermal management device **10** is employed to cool a component of an aircraft, the amount of working fluid **30** may be sufficient to sustain operation of the evaporative thermal management device **10** for the duration of a single trip or for multiple trips. The amount of working fluid **30** in the cavity **22** may be adjusted based on the rate of heat transfer and/or the desired duration of use. Thus, the amount of working

fluid **30** charged into the cavity **22** may be sufficient to accommodate particular heat transfer rate (i.e., to dissipate a particular heat load) for the substrate **12** over a particular period of time.

(35) FIGS. **1-3** illustrate exemplary substrate **12** configurations where the substrate(s) **12** are in thermal communication with the evaporative thermal management device **10**. In FIG. **1**, the substrate **12** is in contact with the top surface of the evaporative thermal management device **10**. In such a configuration, the housing **14** acts as both a mechanical support and a heat transfer device for the substrate **12**. In FIG. **2**, the substrate **12** is in contact with a bottom surface of the evaporative thermal management device **10**. In FIG. **3**, two substrates **12** are in thermal communication with the evaporative thermal management device **10**. One substrate **12** is in contact with the top surface and another substrate **12** is in contact with the bottom surface of the evaporative thermal management device **10**. In some embodiments, one or more substrates **12** may be surrounded by the housing **14** or positioned within an opening in the housing **14**.

(36) Turning to FIGS. **4** and **5**, the evaporative thermal management device **10** further includes at least one porous media domain. A porous media is used to create a porous media domain. The porous media may be any suitable porous structure such as, for example, a mesh structure, a gel structure, or combinations thereof. In FIG. **4**, a fine porous media **35A** is disposed in the cavity **22** of the housing **14**. In FIG. **5**, the fine porous media **35A** and a coarse porous media **35B** are disposed in the cavity **22**. The porous media **35A**, **35B** may be attached to an interior surface of the housing **14** using any suitable attachment mechanisms. For example, the porous media **35A**, **35B** may be attached to the interior surface of the housing **14** by heat sealing, welding, brazing, adhesives, or any mechanical means. In some approaches, the ends of the porous media **35A**, **35B** may be hot dipped in brazing metal to enhance the thermal and mechanical contact with the housing **14**.

(37) The size of the pores in the porous media **35A**, **35B** may vary depending on the particular application or use of the evaporative thermal management device **10**. A finer porous media, such as the porous media **35A**, slows vapor removal from the cavity **22** and provides more containment of the working fluid **30**, preventing movement of working fluid **30** within the cavity **22**. A coarser porous media, such as the coarse porous media **35B**, may speed up vapor removal and provides less containment of the working fluid **30**, permitting more movement of the working fluid **30** within the cavity **22**. The fine porous media **35A** may be employed, for example, in an application in which the evaporative thermal management device **10** is subjected to a large amount of movement or to gravitational (“G”) forces. By contrast, the coarse porous media **35B** may be employed, for example, in an application subjected to a small amount of movement or G-forces. The pore size and/or density of the porous media **35A**, **35B** may also contribute to the rate of heat transfer from the substrate **12** to the working fluid **30**. A denser porous media expedites the rate of heat transfer.

(38) As illustrated in FIGS. **4** and **5**, the porous media **35A**, **35B** confines the working fluid **30** to the base **24** of the housing **14**. While the porous media **35A**, **35B** are disposed along the base **24** of the cavity **22** in FIGS. **4** and **5**, it is contemplated that the porous media may be disposed in any portion of the cavity **22**. In this manner, the porous media **35A**, **35B** may be used to confine the working fluid **30** to any portion of the cavity **22** in which the porous media **35A**, **35B** is positioned. In some embodiments, the porous media **35A**, **35B** is positioned to prevent or minimize the loss of working fluid **30** through the vents **28** prior to evaporation.

(39) Turning to FIG. **6**, an exploded view of an exemplary evaporative thermal management device **48** is illustrated. The evaporative thermal management device **48** includes a top case **36**. Vents **40** are formed in the top case **36** and a plurality of supports **38** are coupled to a surface of the top case **36**. In some embodiments, the top case **36** and the supports **38** are formed of a single piece of material. The evaporative thermal management device **48** also includes a base case **46**. A first porous media **42** and a second porous media **44** are disposed in the interior of the evaporative thermal management device **48**. The first porous media **42** and the second porous media **44** are positioned between the top case **36** and the base case **46**. In some aspects, the first porous media **42**

and the second porous media **44** are mesh layers, however, it is contemplated that other suitable porous structures such as gel structures may be used.

(40) The top case **36** is coupled to the base case **46**. Additionally, the first porous media **42** and the second porous media **44** are coupled to the base case **46**. The top case **36** may be coupled to the base case **46**, and the first porous media **42** and the second porous media **44** attached to the base case **46**, by any suitable attachment mechanism **50**. Suitable attachment mechanisms **50** may include brazing, welding, adhesives, or mechanical means. When assembled, the first porous media **42** forms a partial containment lid for the evaporative thermal management device.

(41) In some approaches, the first porous media **42** and the second porous media **44** are mesh layers. It is contemplated that the mesh layers may perform the same function as the porous media **35A**, **35B** described with reference to FIGS. **4** and **5**.

(42) It is contemplated that the top case **36**, the base case **46**, or portions thereof may be formed using any suitable manufacturing method. Suitable manufacturing methods may include, for example, additive manufacturing approaches such as 3D printing.

(43) With reference to FIG. **7**, an alternative embodiment of a housing **52** for the evaporative thermal management device **10** is illustrated. The housing **52** includes vents **56**. The vents **56** are formed in the top portion of the housing **52**. Vent tubes **58** are coupled to the vents **56**. In FIG. **8**, the vent tubes **58** are shown in an open configuration and without any pressure regulating devices coupled thereto. However, it is also contemplated that, in some embodiments, the vent tubes **58** may include pressure regulating devices, such as the pressure relief valve **92** and venturi pump **94** described with reference to FIGS. **14** and **15**.

(44) The housing **52** includes a plurality of branched supports **54** disposed in the cavity **60**. The branched supports **54** include a forked distribution of support posts that create a porous domain in the bottom of the cavity **60**. The branched supports **54** include wider spaced posts at the top and narrower spaced posts at the bottom. In this manner, the branched supports **54** create small pores **64** towards the bottom of the cavity **60** and large pores **62** towards the top of the cavity **60**. The forked configuration of the branched supports **54** increases the rate of heat transfer through the branched supports **54** and also confines the working fluid to the bottom portion of the cavity **60**. The branched supports **54** may also provide additional structural support for a substrate positioned on the top of the housing **52**.

(45) With reference to FIGS. **8-10**, another exemplary evaporative thermal management device **66** is illustrated. The evaporative thermal management device **66** includes a housing **68** that is in thermal communication with a substrate **70**. The housing **68** includes plurality of supports **74**. The supports **74** are cylindrical columns that are arranged in rows within the housing **68**. Although the supports **74** are depicted in rows in FIGS. **8-10**, the supports **74** may be positioned in any arrangement suitable to support the substrate **70**, to provide a suitable surface area to volume ratio of the housing, to achieve a suitable heat transfer rate, and/or to provide another mechanical or structural function. The housing **68** also defines cavity **76** with a working fluid **78** disposed therein. A plurality of vent tubes **72** are coupled to the housing **68**. The vent tubes **72** allow evaporated working fluid **78** to flow out of the cavity **76** to the ambient environment.

(46) With reference to FIGS. **11-13**, the evaporative thermal management device **66** is illustrated with a variety of porous media disposed in the cavity **76**. The porous media in the cavity **76** confines the working fluid **78** to a particular portion of the cavity **76**. The porous media provides internal containment and helps to hold the working fluid domains in place within the cavity **76**.

(47) In FIG. **11**, the cavity **76** includes a porous mesh structure **86** as the porous media. The porous mesh structure **86** is disposed along a floor **80** of the cavity **76** and fills a bottom portion of the cavity **76**. The working fluid **78** may fill the porous mesh structure **86** or a portion thereof. In operation, the working fluid **78** is heated by the housing **68**. The porous mesh structure **86** further conducts heat from the housing **68** through the working fluid **78**. The heated working fluid **78** evaporates and travels through the porous mesh structure **86** and exits through the vent tubes **72** to

the ambient environment.

(48) In FIG. 12, the cavity 76 includes a porous mesh structure 88 as the porous media. The porous mesh structure 88 creates a porous media domain. The porous mesh structure 88 is disposed along the floor 80 of the cavity 76 and fills the bottom portion of the cavity 76. The porous mesh structure 88 also lines the walls 82 and the ceiling 84 of the cavity 76. The working fluid 78 may fill the porous mesh structure 88 or a portion thereof. In operation, the working fluid 78 is heated by the housing 68 and the porous mesh structure 88 further conducts heat from the housing through the working fluid 78. The heated working fluid 78 evaporates and travels through the porous mesh structure 88 and exits through the vent tubes 72 to the ambient environment.

(49) Further, the porous mesh structure 88 distributes the working fluid 78 around the interior perimeter of the cavity, increasing the surface area of the cavity 76 that is contact with working fluid 78. The porous mesh structure 88 also covers the vent tube 72 opening to prevent working fluid 78 that is in the liquid state from escaping through the vent tube 72. When positioned in the manner, the porous mesh structure 88 also allows evaporated working fluid 78 that is in the gaseous state to exit through the vent tube 72. Capillary forces in the porous mesh structure 88 help to keep liquid working fluid 78 in the housing 68.

(50) In addition to including a porous media to contain the working fluid 78 within the housing 68, one or more vent tubes 72 may include a structure or device to prevent non-evaporated working fluid 78 that is in the liquid state from escaping through the vents. In some approaches, one or more of the vents on any of the evaporative thermal management device described herein may include a vapor-liquid separator to remove liquids entrained in evaporated working fluid and/or to redirect the liquid back into the housing. In some aspects, the vapor-liquid separator may be a centrifugal-type separator. In some aspects, the vapor-liquid separator may include one or more baffles or impingement plates disposed in the vent. The vapor-liquid separator may also use a porous wick structure to collect liquid droplets and return them to the housing.

(51) In FIG. 13, the evaporative thermal management device 66 includes a vapor-liquid separator 91. The vapor-liquid separator 91 may be separate from or integral with the housing 68, for example, the housing 68 may morph into the vapor liquid-separator 91. The vapor-liquid separator 91 may be any suitable vapor-liquid separation device or structure. The vapor-liquid separator 91 is coupled to a vent opening 93 on the housing 68. The vapor-liquid separator 91 includes baffles 97 that define a vent channel 95. The baffles 97 create a tortuous path through the vapor-liquid separator 91. The porous mesh structure 88 extends from the cavity 76 of the housing 68 across the vent channel 95.

(52) In operation, the substrate 70 heats the working fluid 78 which causes the working fluid 78 to evaporate. Evaporated working fluid 78 exits the housing 68 through the vent tube 72 and the vapor-liquid separator 91. The evaporated working fluid 78 may include entrained liquids. The vapor-liquid separator 91 removes entrained liquids from the exiting gas stream. The porous mesh structure 88 acts as a wick structure to collect any liquid droplets that hit it and directs the liquids back into the housing 68.

(53) Further, changes in the orientation of the evaporative thermal management device 66 may cause the vent tube 72 to become submerged in liquid working fluid 78. Submersion or other blockages in the vent tube 72 may prevent evaporated working fluid 78 from exiting. When evaporated working fluid 78 is trapped in the housing 68, the housing 68 may pressurize and force both evaporated and liquid working fluid 78 out of the vent opening 93. The vapor-liquid separator 91 deflects liquid working fluid 78 back into the housing 68 and remove liquid working fluid 78 from the exiting gas stream.

(54) In FIG. 14, the cavity 76 includes a gel structure 90 as the porous media. The gel structure 90 creates a porous media domain. It is contemplated that the gel structure 90 may perform the same function as the porous media 35A, 35B described with reference to FIGS. 4 and 5. The gel structure 90 is disposed along the floor 80 of the cavity 76 and fills a bottom portion of the cavity 76. The

working fluid **78** may fill the gel structure **90** or a portion thereof. In operation, the working fluid **78** is heated by the housing **68** and the gel structure **90** further conducts heat from the housing through the working fluid **78**. The heated working fluid **78** evaporates and travels through the gel structure **90** and exits through the vent tubes **72** to the ambient environment.

(55) Turning to FIG. **15**, the evaporative thermal management device **66** includes pressure relief valves **92** on the vent tubes **72**. The pressure relief valves **92** allow for tuning of the saturation pressure and, therefore tuning of the boiling point or saturation temperature, of the working fluid **78**. Using the pressure relief valves **92**, the evaporative thermal management device **66** maintains the pressure of the cavity **76** at or below the set point of the pressure relief valves **92**.

(56) Turning to FIG. **16**, the evaporative thermal management device **66** includes venturi pumps **94** on the vent tubes **72**. The venturi pumps **94** control the pressure of the cavity **76**. In particular, the venturi pumps **94** blow motive air across the vent tubes **72** to actively lower the pressure of the cavity **76**. Accordingly, the pressure of the cavity **76** may be adjusted by the flow rate of the motive air. By lowering the operating pressure of the cavity **76**, the venturi pumps **94** may also lower the boiling point of the working fluid **78** that is held in the cavity **76**. In a low-pressure environment, the working fluid **78** is able to evaporate at a lower temperature and maintains the substrate **70** at a lower temperature.

(57) Although pressure relief valves **92** and venturi pumps **94** are shown as the pressure regulating devices in FIGS. **15** and **16**, it is contemplated that any number of suitable pressure regulating devices may be used to control the pressure in the cavity **76**.

(58) FIGS. **17** and **18** illustrate additional housing configurations for the evaporative thermal management devices described herein. FIG. **17** illustrates a non-planar housing configuration, and FIG. **18** illustrates an angled housing configuration. However, it is to be understood that the evaporative thermal management devices described herein may employ other housing configurations to accommodate the shape of a particular substrate or to accommodate the space limitations of a particular cooling application.

(59) FIG. **17** depicts an evaporative thermal management device **100** having a non-planar housing **104**. The non-planar housing **104** is in thermal communication with a plurality of substrates **102**. The non-planar housing **104** comprises an arcuate base case **108** and an arcuate top case **110**. The non-planar housing **104** further includes a plurality of supports **112**, which extend from the arcuate base case **108** to the arcuate top case **110** and form a cavity **116** therebetween. The cavity **116** holds a working fluid **126**. A porous structure **124** lines the floor **118**, the walls **120**, and the ceiling **122** of the cavity **116**. The porous structure **124** forms a porous domain within the cavity **116**. The porous structure **124** helps to confine the working fluid **126** to the floor **118**, the walls **120**, and the ceiling **122** of the cavity **116**. Two vents **114** are disposed at end portions of the non-planar housing **104**. The vents **114** are in fluid communication with the environment surrounding the non-planar housing **104**. The vents **114** are configured to release evaporated working fluid **126** from the non-planar housing **104** during operation of the evaporative thermal management device **100**.

(60) FIG. **18** depicts an evaporative thermal management device **130** having an angled housing **134**. The angled housing **134** is formed in an L-shape. The non-planar housing **104** is in thermal communication with substrates **132**. The angled housing **134** comprises an angled base case **138** and an angled top case **140**. The angled housing **134** further includes a plurality of supports **142**, which extend from the angled base case **138** to the angled top case **140** and form a cavity **146** therebetween. The cavity **146** holds a working fluid **156**. A porous structure **154** lines the floor **148**, walls **150**, and ceiling **152** of the cavity **146**. The porous structure **154** forms a porous domain within the cavity **146**. The porous structure **154** helps to confine the working fluid **156** to the floor **148**, the walls **150**, and the ceiling **152** of the cavity **146**. Two vents **144** are disposed at end portions of the angled housing **134**.

(61) The vents **144** each include a valve **158**. In some approaches, the valves **158** are squib valves. The valves **158** may be configured place the vents **144** in fluid communication with the

environment surrounding the angled housing **134**. In some approaches, the valves **158** may open to initiate operation of the evaporative thermal management device **130**. When the vents **144** are in fluid communication with the environment surrounding the angled housing **134**, the vents **144** release evaporated working fluid **156** from the angled housing **134** and expose the cavity **146** to the pressure of the environment surrounding the angled housing **134**.

(62) In some embodiments, in any of the evaporative thermal management devices described here, instead of venting the evaporated working fluids to the ambient environment, the evaporated working fluids may be used in one or more secondary cooling applications. In this manner, the evaporated working fluid may be repurposed and used as a cooling fluid for other secondary substrates. The evaporated working fluid may be separated and/or collected from the vents of the evaporative thermal management device and directed another substrate or system. The vents of the evaporative thermal management devices may be directly coupled to the secondary substrate or system.

(63) In some approaches, the vents of the thermal management devices described herein may be coupled to additional tubes, pipes, or other conduits to direct the evaporated working fluid to the secondary substrate or system. In some examples, the tubes, pipes, or other conduits may be disposed inside of a cold plate. The cold plate may be in thermal communication with the secondary substrate or system. In this manner, the evaporated working fluid flowing through the tubes, pipes, or other conduits may be used to cool the secondary substrate or system.

(64) Turning to FIG. **19**, in one approach, an evaporative thermal management device **160** is coupled to a secondary system **168**. The evaporative thermal management device **160** may be any one of the evaporative thermal management devices described herein. The evaporative thermal management device **160** is in thermal communication with a primary substrate **162**. The primary substrate **162** may be any heat producing substrate. The evaporative thermal management device **160** includes vents **164**. The vents **164** are coupled to tubes **166** and collect evaporated working fluid from the evaporative thermal management device **160** and direct the evaporate working fluid to the secondary system **168**. The secondary system **168** includes a cold plate **172** and a secondary substrate **170** that is in thermal communication with the cold plate **172**. The secondary substrate **170** may be any heat producing substrate.

(65) In operation, the evaporative thermal management device **160** cools the primary substrate **162**. Heat from the primary substrate **162** heats and evaporates a working fluid that is housed within the evaporative thermal management device **160**. The evaporated working fluid exits the evaporative thermal management device **160** through the vents **164**. The evaporated working fluid then flows from the vents **164** into tubes **166**. The tubes **166** direct the evaporated working fluid through the cold plate **172** to cool the secondary substrate **170**.

(66) Applications of the evaporative thermal management devices described herein are illustrated in the following examples.

Example 1

(67) In one exemplary application, the evaporative thermal management device **10** of FIG. **1** is used to maintain a substrate **12** at or below a threshold temperature of 70° C. The substrate **12** is placed in thermal communication contact with the evaporative thermal management device **10**. No pressure regulating device is employed on the vents **28** of the evaporative thermal management device **10**. Therefore, when the evaporative thermal management device **10** is used at sea level, the operating pressure of the evaporative thermal management device **10** is about 1 atm. To maintain the substrate **12** at or below the threshold temperature of 70° C., methanol is used as the working fluid **30**. The boiling point of methanol at a pressure of 1 atm is about 65° C. In operation, when the vents **28** are open, the operating pressure of the device is brought to 1 atm and the methanol evaporates, maintaining the temperature of the substrate **12** at about 65° C.

Example 2

(68) In another exemplary application, the evaporative thermal management device **10** of FIG. **1** is

used to maintain a substrate **12** at or below a threshold temperature of 75° C. The substrate **12** is placed in thermal communication contact with the evaporative thermal management device **10**. No pressure regulating device is employed on the vents **28** of the evaporative thermal management device **10**. Therefore, when the evaporative thermal management device is used at about 30,000 feet above sea level, the operating pressure of the thermal management device is about 0.29 atm. To maintain the substrate **12** at or below the threshold temperature of 75° C., water is used as the working fluid **30**. The boiling point of water at a pressure of 0.29 atm is about 71° C. In operation, when vents **28** are open, the operating pressure of the evaporative thermal management device **10** is brought to about 0.29 atm and the water evaporates, maintaining the temperature of the substrate at about 71° C.

Example 3

(69) In another exemplary application, the evaporative thermal management device **66** of FIG. **13** is used to maintain a substrate **70** at or below a threshold temperature of about 82° C. The evaporative thermal management device **66** employs water as the working fluid **78**. To maintain the substrate at the threshold temperature of about 82° C., the venturi pump **94** is used to adjust the pressure of the cavity **76** at about 0.5 atm. By lowering the operating pressure of the evaporative thermal management device **66** to about 0.5 atm, the venturi pump **94** lowers the boiling point of water to about 82° C. The temperature of the substrate **70** is therefore maintained at about 82° C.

Example 4

(70) In another exemplary application, the pressure regulating devices in the evaporative thermal management device **66** are pressure relief valves **92** (see FIG. **15**). The pressure relief valves provide a constant interior pressure and therefore a constant working fluid **78** temperature as exterior ambient pressure changes, for example, due to changes in altitude.

(71) Further aspects of the invention are provided by the subject matter of the following clauses:

(72) An evaporative thermal management device comprising: a housing having a base case; a top case positioned opposite the base case; a plurality of supports extending from the base case to the top case and forming a cavity therebetween, the cavity having an operating pressure associated therewith; at least one vent coupled to the housing and in fluid communication with the cavity; and a working fluid disposed in the cavity, the working fluid having a boiling point at or below a threshold temperature associated with a substrate at the operating pressure; wherein evaporation of the working fluid controls a temperature of the substrate when the cavity is at the operating pressure, and the evaporated working fluid is vented to an ambient environment surrounding the housing through the at least one vent.

(73) The evaporative thermal management device of any preceding clause, wherein the at least one vent includes a pressure regulating device that controls a pressure within the housing to adjust the boiling point of the working fluid.

(74) The evaporative thermal management device of any preceding clause, wherein the at least one vent includes a vapor-liquid separator.

(75) The evaporative thermal management device of any preceding clause, wherein at least one of the plurality of supports is a column.

(76) The evaporative thermal management device of any preceding clause, wherein the support has a branched structure comprising a plurality of posts.

(77) The evaporative thermal management device of any preceding clause further comprising: a porous domain disposed within the cavity for containment of the working fluid within the cavity.

(78) The evaporative thermal management device of any preceding clause, wherein the porous domain is at least one of: a mesh and a gel.

(79) The evaporative thermal management device of any preceding clause, wherein the porous domain is disposed on a bottom wall of the cavity.

(80) The evaporative thermal management device of any preceding clause, wherein the porous domain is further disposed on one or more side walls of the cavity.

- (81) The evaporative thermal management device of any preceding clause, wherein the at least one vent includes a valve configured to open the at least one vent and place the at least one vent in fluid communication with the ambient environment.
- (82) The evaporative thermal management device of any preceding clause, wherein the operating pressure is a pressure of the ambient environment.
- (83) An evaporative thermal management system comprising: a substrate having a threshold temperature associated therewith; an evaporative thermal management device in thermal communication with the substrate, the evaporative thermal management device comprising: a housing having a base case, a top case, and a plurality of supports extending from the base case to the top case and forming a cavity therebetween, the cavity having an operating pressure associated therewith; at least one vent coupled to the housing and in fluid communication with the cavity; and a working fluid disposed in the cavity, the working fluid having a boiling point at or below a threshold temperature associated with a substrate at the operating pressure, wherein evaporation of the working fluid controls the temperature of the substrate when the cavity is at the operating pressure, and the evaporated working fluid is vented to an ambient environment surrounding the housing through the at least one vent.
- (84) The evaporative thermal management system of any preceding clause, wherein the plurality of supports are sized to mechanically support the weight of the substrate.
- (85) The evaporative thermal management system of any preceding clause, wherein the top case of the housing is in thermal communication with the substrate.
- (86) The evaporative thermal management system of any preceding clause, wherein the bottom case of the housing is in thermal communication with the substrate.
- (87) A method of thermal management for a substrate using an evaporative thermal management device, the evaporative thermal management device having a housing with a base case, a top case positioned opposite the base case, a plurality of supports extending from the base case to the top case and forming a cavity therebetween, and at least one vent coupled to the housing and in fluid communication with the cavity, the method comprising: placing the evaporative thermal management device in thermal communication with the substrate; associating an operating pressure with the cavity; filling at least a portion of the cavity with a working fluid, the working fluid having a boiling point at or below a threshold temperature associated with the substrate at the operating pressure; exposing the cavity to the operating pressure; evaporating the working fluid to control the temperature of the substrate; and venting the evaporated working fluid to an ambient environment surrounding the housing via the at least one vent.
- (88) The method of any preceding clause further comprising the step of: controlling the boiling point of the working fluid via a pressure regulating device on the at least one vent.
- (89) The method of claim any preceding clause, wherein the step of exposing further comprises adjusting the pressure of the cavity using the pressure regulating device.
- (90) The method of any preceding clause further comprising the step of: opening a valve on the at least one vent to place the cavity in fluid communication with the ambient environment.
- (91) The method of any preceding clause, further comprising the step of: using the evaporated working fluid in a secondary cooling application.
- (92) It will be understood that various changes in the details, materials, and arrangements of parts and components which have been herein described and illustrated to explain the nature of the dynamic seals between moving components and stationary components may be made by those skilled in the art within the principle and scope of the appended claims. Furthermore, while various features have been described with regard to particular embodiments, it will be appreciated that features described for one embodiment also may be incorporated with the other described embodiments.

Claims

1. An evaporative thermal management device comprising: a moveable housing having a wall with an exterior surface and an interior surface, the interior surface defining at least in part an interior cavity; at least one non-wicking support extending across at least a portion of the interior cavity to connect a first portion of the interior surface to a separate second portion of the interior surface and engaging at least a portion of the interior surface to support at least a portion of the wall to stiffen the moveable housing from within the moveable housing; at least one vent coupled in fluid communication with the interior cavity; and a fluid disposed in the interior cavity, the fluid being free to move about the interior cavity as the moveable housing moves; wherein the exterior surface contacts a substrate, evaporation of the fluid controls a temperature of the substrate, the evaporated fluid forming in a space in the interior cavity between the fluid and the interior surface is vented to an ambient environment surrounding the moveable housing through the at least one vent.
2. The evaporative thermal management device of claim 1, wherein the moveable housing is non-rectangular.
3. The evaporative thermal management device of claim 1, wherein the moveable housing includes a curved portion.
4. The evaporative thermal management device of claim 1, wherein the at least one vent includes a pressure regulator that controls a pressure within the moveable housing to adjust a boiling point of the fluid.
5. The evaporative thermal management device of claim 1, wherein the at least one vent includes a vapor-liquid separator that returns liquid to the interior cavity.
6. The evaporative thermal management device of claim 5, wherein the vapor-liquid separator comprises at least one baffle.
7. The evaporative thermal management device of claim 6, wherein the at least one baffle defines a tortuous path through the vapor-liquid separator.
8. The evaporative thermal management device of claim 5, wherein the vapor-liquid separator comprises a porous media that entrains liquid and returns the entrained liquid to the interior cavity.
9. The evaporative thermal management device of claim 1, wherein the at least one non-wicking support has a branched structure comprising a plurality of posts.
10. The evaporative thermal management device of claim 1, wherein the at least one vent includes a valve configured to open the at least one vent and place the at least one vent in fluid communication with the ambient environment.
11. An evaporative thermal management device comprising: a housing having a wall with an exterior surface and an interior surface, the interior surface defining at least in part an interior cavity; at least one support extending across at least a portion of the interior cavity to connect a first portion of the interior surface to a separate second portion of the interior surface to support at least a portion of the exterior surface; at least one vent coupled in fluid communication with the interior cavity; a fluid disposed in the interior cavity; and a porous media disposed in at least a portion of the interior cavity to limit movement of the fluid in the interior cavity, the at least one support extending through at least a portion of the porous media having a level of porosity, the level of porosity being determined by an amount of movement force to be experienced by the housing, wherein the exterior surface contacts a substrate, evaporation of the fluid controls a temperature of the substrate, the evaporated fluid forming in a space in the interior cavity between the fluid and the interior surface is vented to an ambient environment surrounding the housing through the at least one vent of the at least two vents.
12. The evaporative thermal management device of claim 11, wherein the porous media is at least one of a mesh or a gel.
13. The evaporative thermal management device of claim 12, wherein the porous media is

positioned at the at least one vent.

14. The evaporative thermal management device of claim 11, wherein the porous media includes a first porous media of a first level of porosity and a second porous media of a second level of porosity, the first level of porosity being coarser than the second level of porosity.

15. The evaporative thermal management device of claim 14, wherein the first porous media and the second porous media are layered.

16. The evaporative thermal management device of claim 11, wherein the porous media includes brazen metal attaching the porous media to the interior surface of the interior cavity.

17. The evaporative thermal management device of claim 11, wherein the porous media covers at least a portion of the interior surface of the interior cavity.

18. The evaporative thermal management device of claim 11, wherein the at least one vent includes a vapor-liquid separator that returns liquid to the interior cavity.

19. An evaporative thermal management device comprising: a housing having a wall with an exterior surface and an interior surface, the interior surface defining at least in part an interior cavity; at least one non-wicking support extending across at least a portion of the interior cavity to connect a first portion of the interior surface to a separate second portion of the interior surface and engaging at least a portion of the interior surface to support at least a portion of the wall; at least two vents coupled in fluid communication with the interior cavity; and a fluid disposed in the interior cavity; wherein the exterior surface contacts a substrate, evaporation of the fluid controls a temperature of the substrate, the evaporated fluid forming in a space in the interior cavity between the fluid and the interior surface is vented to an ambient environment surrounding the housing through the at least one vent of the at least two vents.

20. The evaporative thermal management device of claim 19, wherein the at least two vents are spaced from one another such that when fluid in the interior cavity occludes one vent another vent remains open.
